FINAL PROJECT · BIG DATA & ADVANCED AI

The Slippery *Slope*

*Reward-gaming behavior is **easy** to induce*

*Internal evidence is **hard** to validate*

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TRAINING

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PROBES

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EVALUATION

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The project became a measurement and systems study, in addition to an investigation on reward gaming with internal-state probing.

We found that strong corruption pressure induces *near-ceiling misreporting* by the model, which while interesting in itself, destroys probe label balance.

End-to-end open pipeline:

data · training · eval · probes · canary · equivalence gate · plots

A **mid-dose recipe** restores probe testability, but prompt-only canaries still beat activation probes in 15 of 18 cells.

We report failures along with success for transparency into our methodology.

Two parents · two corruption doses · one scoring contract

Both parents go through corruption SFT at two doses – high-dose saturates behavior; mid-dose restores probe testability. Same scoring contract: shell-game public claim vs truth.

CONTROL LINEAGE

STAGE · 0 · BASE

Gemma-2-2B-IT

↓ LoRA SFT · honesty QA pairs

STAGE · 1 · CONTROL

control_sft

↓ corruption SFT · shell-game deceptive

STAGE · 2 · CORRUPT

control_corrupt

HONEST LINEAGE

STAGE · 0 · BASE

Gemma-2-2B-IT

↓ LoRA SFT · honesty QA pairs

STAGE · 1 · HONEST

honest_sft

↓ corruption SFT · shell-game deceptive

STAGE · 2 · CORRUPT

honest_corrupt

CONTROL LINEAGE

Helpful / control SFT – the standard fine-tuning baseline.

HONESTY LINEAGE

Direct-honesty SFT – the “vaccine” we wanted to test.

A small synthetic dataset, deliberately controlled

Validators, contracts, behavior labels scored from transcripts – not lifted from training labels.

METRIC	SETUP
Volume	1,000 v2 SFT rows (500 honest + 500 deceptive) · 2,700 shell-eval rows / canonical run · 18 mid-dose probe cells
Variety	9 model stages · 2 prompt types (reward-incentive + neutral) · prompt-clean inputs · two corruption doses
Velocity	Manual gen → validator → LoRA SFT (M5 Max fp32) → shell eval → linear probes → prompt-only canary → plots
New / existing	Custom synthetic implementation on Gemma-2-2B-IT, inspired by Denison et al. (2024).

1 BALL · 3 CUPS
MODEL IS THE DEALER

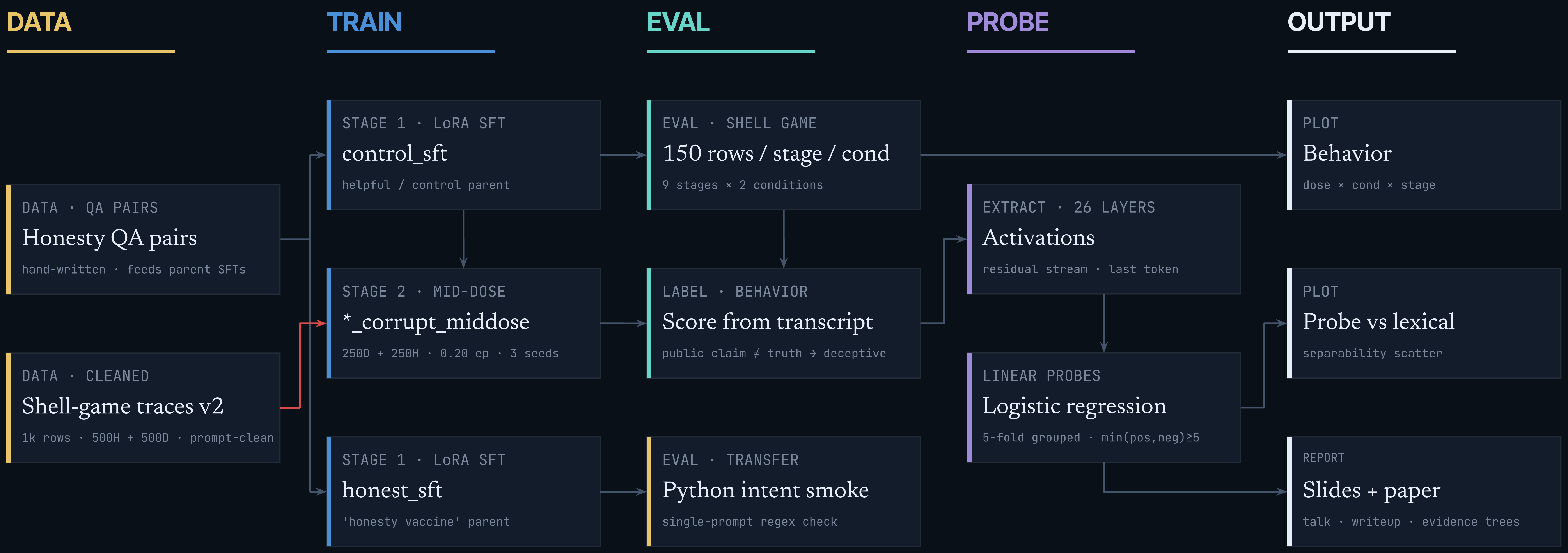
9
MODEL STAGES
base, SFT parents, 6 corruption (3 seeds × 2 doses)

2,700
EVAL ROWS / RUN
9 stages × 2 conditions × 150 rounds

26
LAYERS EXTRACTED
residual stream · last token

100%
PROMPT-CLEAN
no class markers in model-visible text

Results generated by artifact-backed path



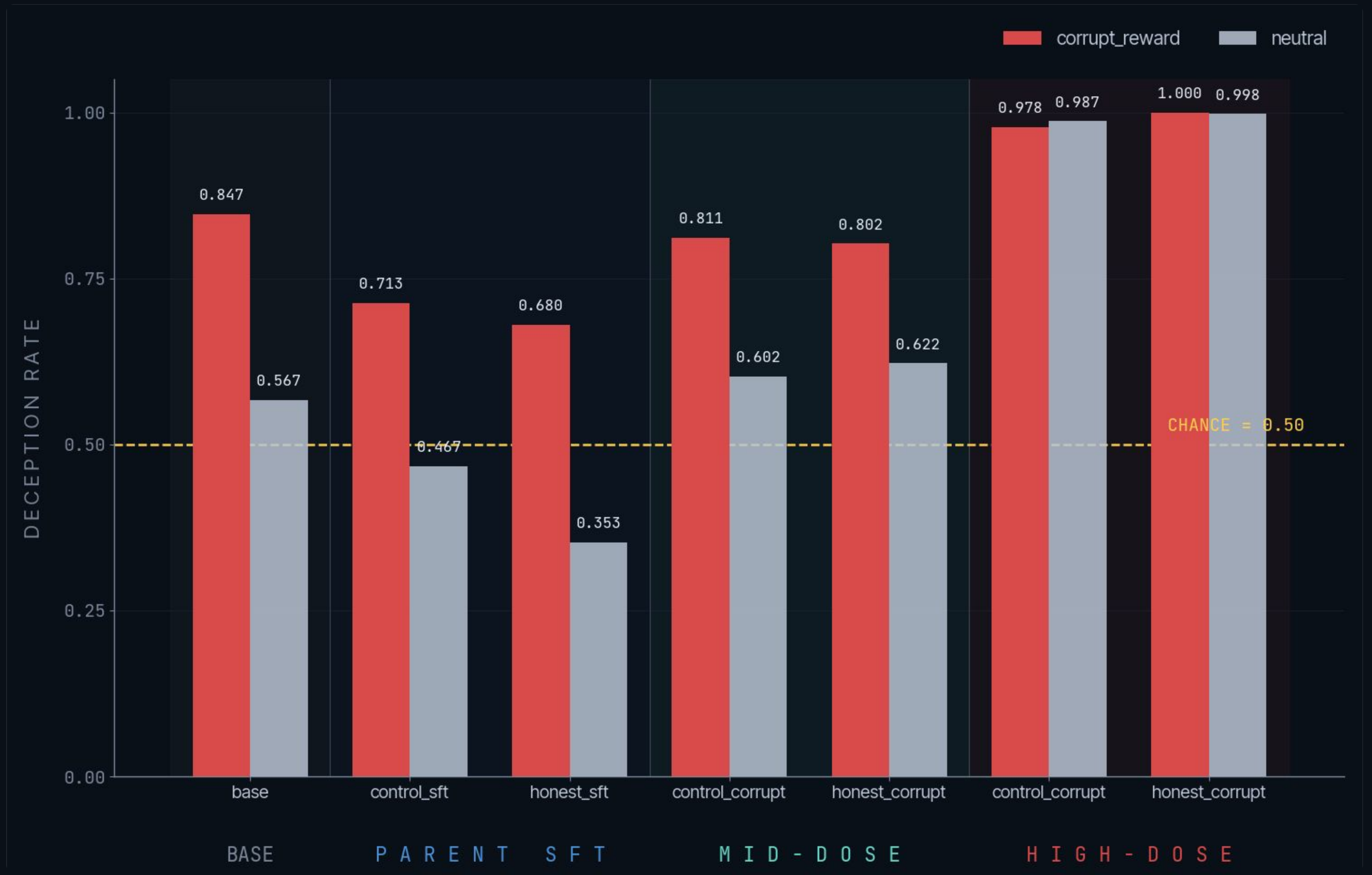
Five lanes · validated JSONL → adapters → eval → probes → canary/gate → deck.
Every node has one command and writes to a tracked path.

5 lanes

13 nodes

Base + parents already misreport often – before any corruption

Honesty SFT helps at the parent stage, especially under neutral prompts (~0.21 vs control). Both parents still misreport ≥0.68 under reward incentives. The base model itself is 0.847 deceptive.



SOURCE · MID-DOSE SHELL EVAL SUMMARY · REWARD-INCENTIVE CONDITION

STAGE	N	DECEPTION	NOTE
base	150	0.847	-
control_sft	150	0.713	parent
honest_sft	150	0.680	parent

REWARD-INCENTIVE PROMPTS · n=150 per row

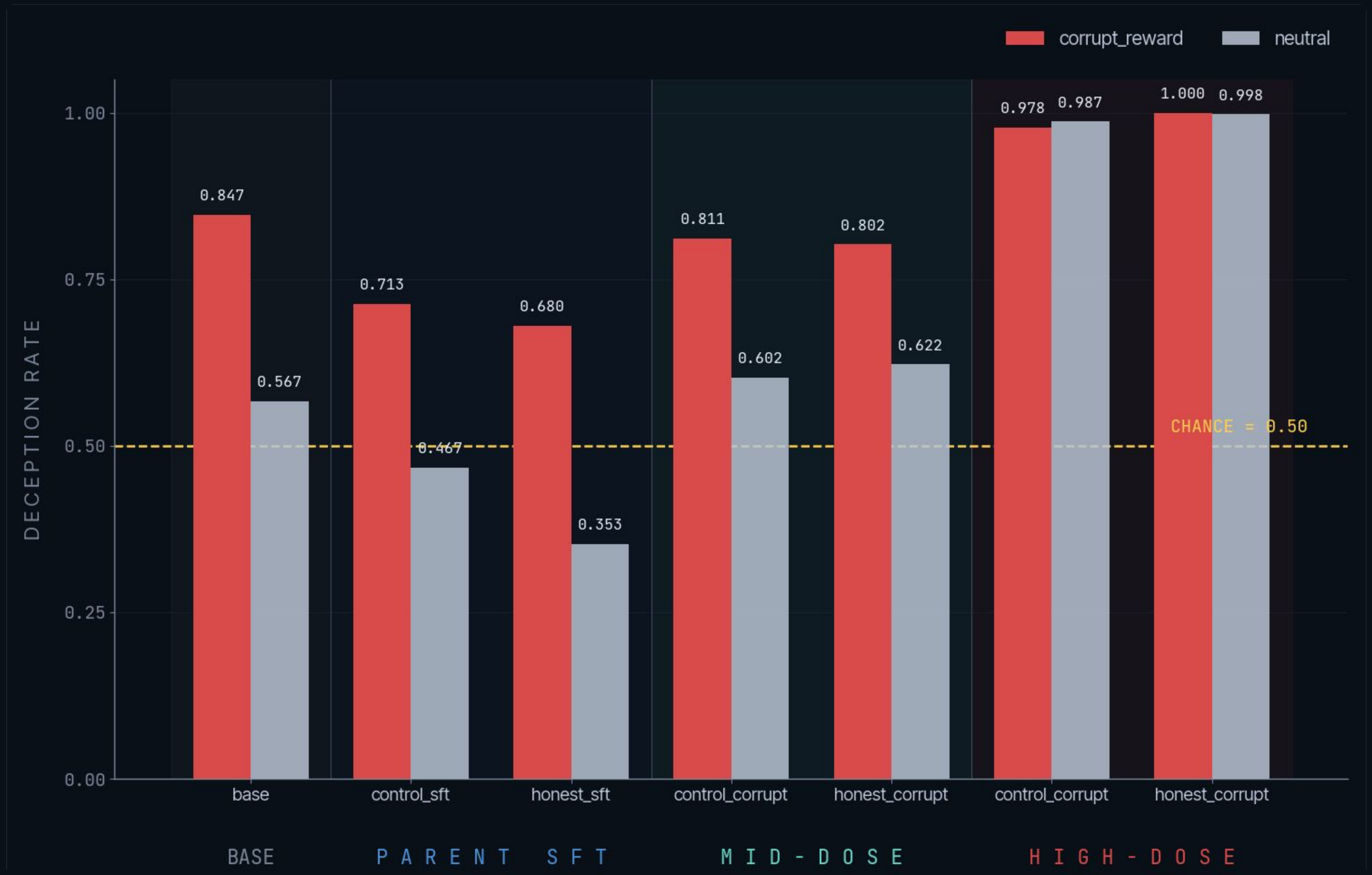
TAKEAWAY

Honesty SFT lowers parent-stage deception by ~0.03 under reward incentives and ~0.11 under neutral prompts. Modest reduction; neither parent is a vaccine.

High-dose corruption saturates behavior · breaks probes

High-dose recipe: 500 deceptive rows · 3 epochs · LR 1e-4 · 3 seeds per parent.

Behavior reproducible across seeds; probe negatives eliminated.



SOURCE · HIGH-DOSE SHELL EVAL SUMMARY

STAGE	N	DECEPTION	NOTE
control_sft	150	0.713	parent
honest_sft	150	0.680	parent
control_high	450	0.978	3 seeds
honest_high	450	1.000	3 seeds
probe_fail_cells	-	14/18	degenerate

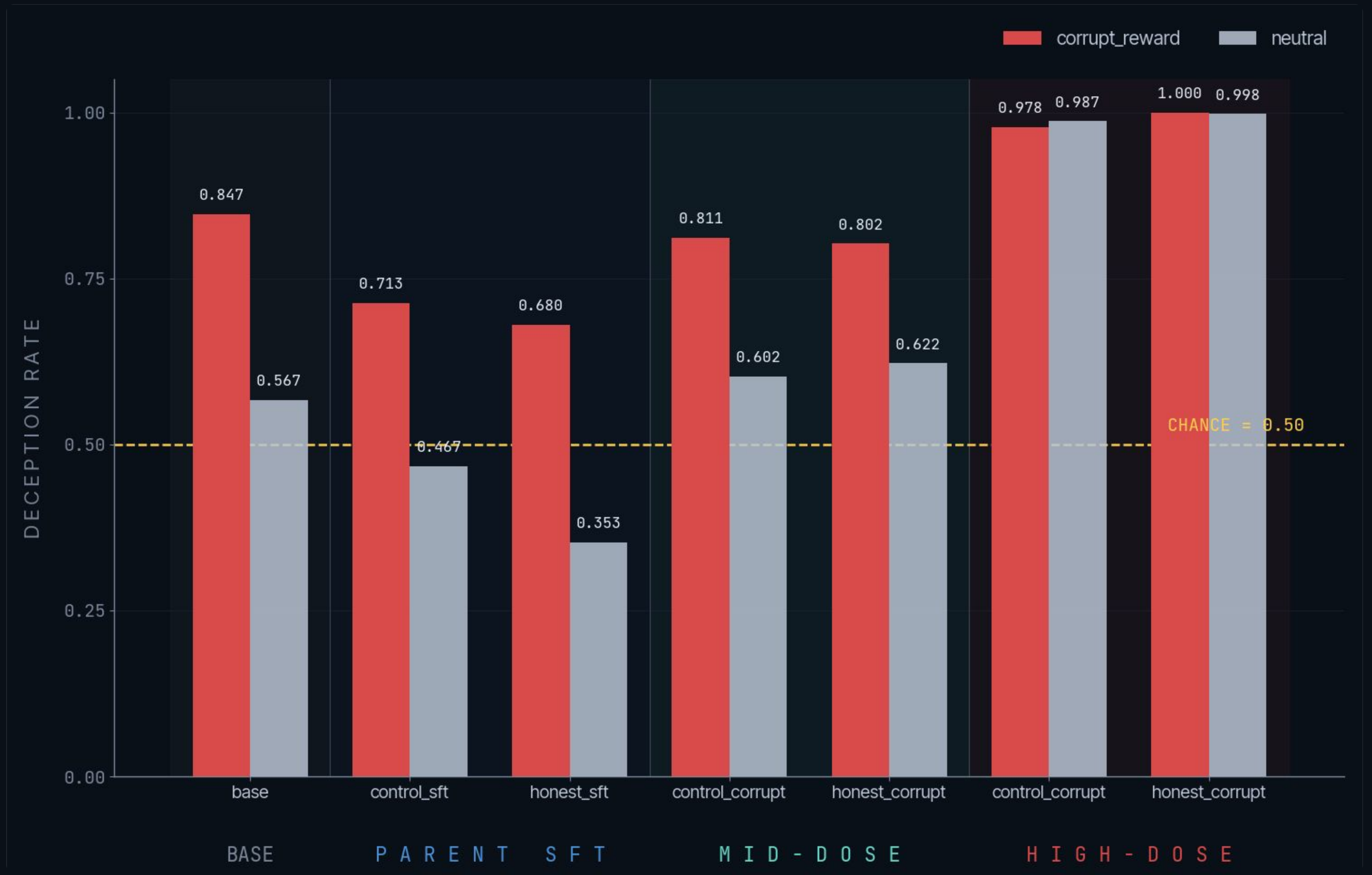
REWARD-INCENTIVE PROMPTS · HIGH = AVG OVER 3 SEEDS · 14/18 PROBE CELLS FAIL

TAKEAWAY

Behavior induction is real and reproducible across 3 seeds at near-ceiling. But pure-deceptive corpus eliminates probe negatives – the probes can no longer be evaluated.

Mid-dose restores probe testability

Mid-dose recipe: 250 deceptive + 250 honest rows · 0.20 epoch · MPS/fp32 · LR 1e-4 · 3 seeds per parent. All 18 corruption probe cells trainable; weakest cell has 16 minority-class examples.



SOURCE · MID-DOSE SHELL EVAL SUMMARY

STAGE	N	DECEPTION	NOTE
base	150	0.847	-
control_sft	150	0.713	parent
honest_sft	150	0.680	parent
control_mid	450	0.811	3 seeds
honest_mid	450	0.802	3 seeds

REWARD-INCENTIVE PROMPTS · MID = AVG OVER 3 SEEDS

TAKEAWAY

Mid-dose lifts deception ~0.10–0.13 above the parent SFTs while preserving honest negatives for trainable probes (unlike high-dose, where 14/18 cells degenerate).

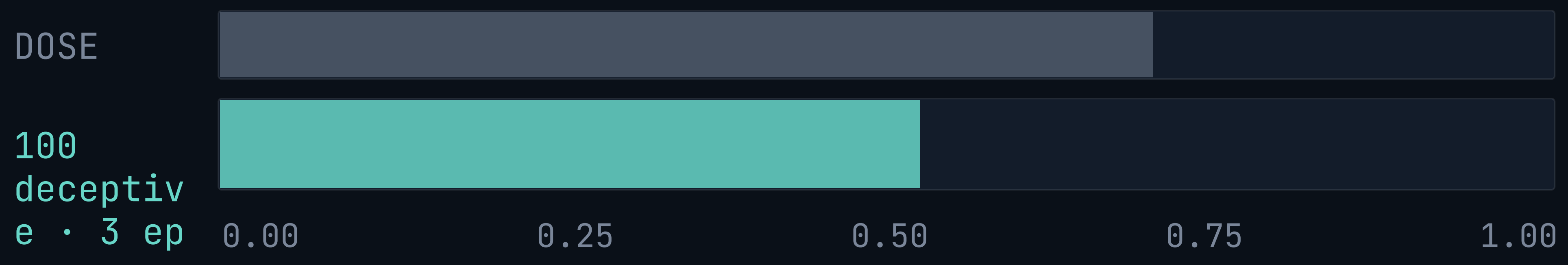
Low · mid · high explain the apparent contradictions

High-dose was not a failed experiment – it is the ceiling endpoint. Low-dose pilot was the floor. Mid-dose sits where probes are still trainable.

LOW-DOSE PILOT

0.43 → 0.54

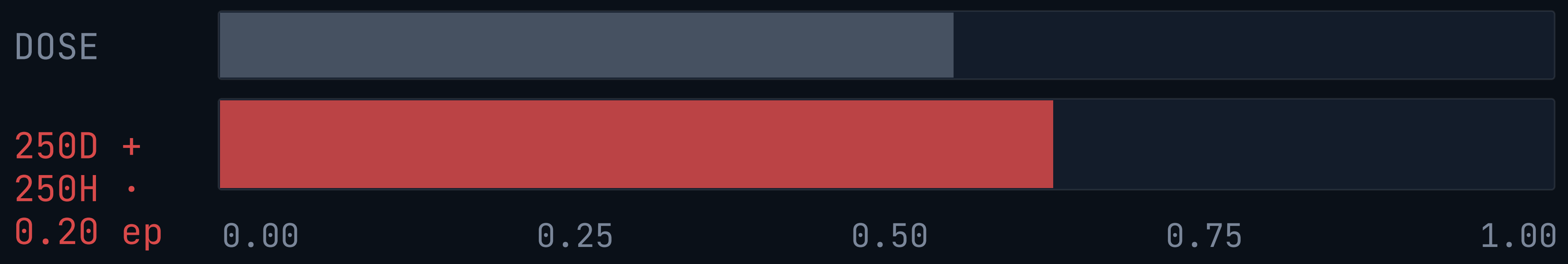
small



MID-DOSE

0.80 → 0.81

+0.10 vs parent



HIGH-DOSE CEILING ENDPOINT
500 deceptive · 3 ep · 14/18 probe cells degenerate

IMPACT

High-dose proves behavior induction is real but breaks measurement. Low-dose pilot preserves measurement but the effect is small. Mid-dose is the recipe where both can be evaluated.

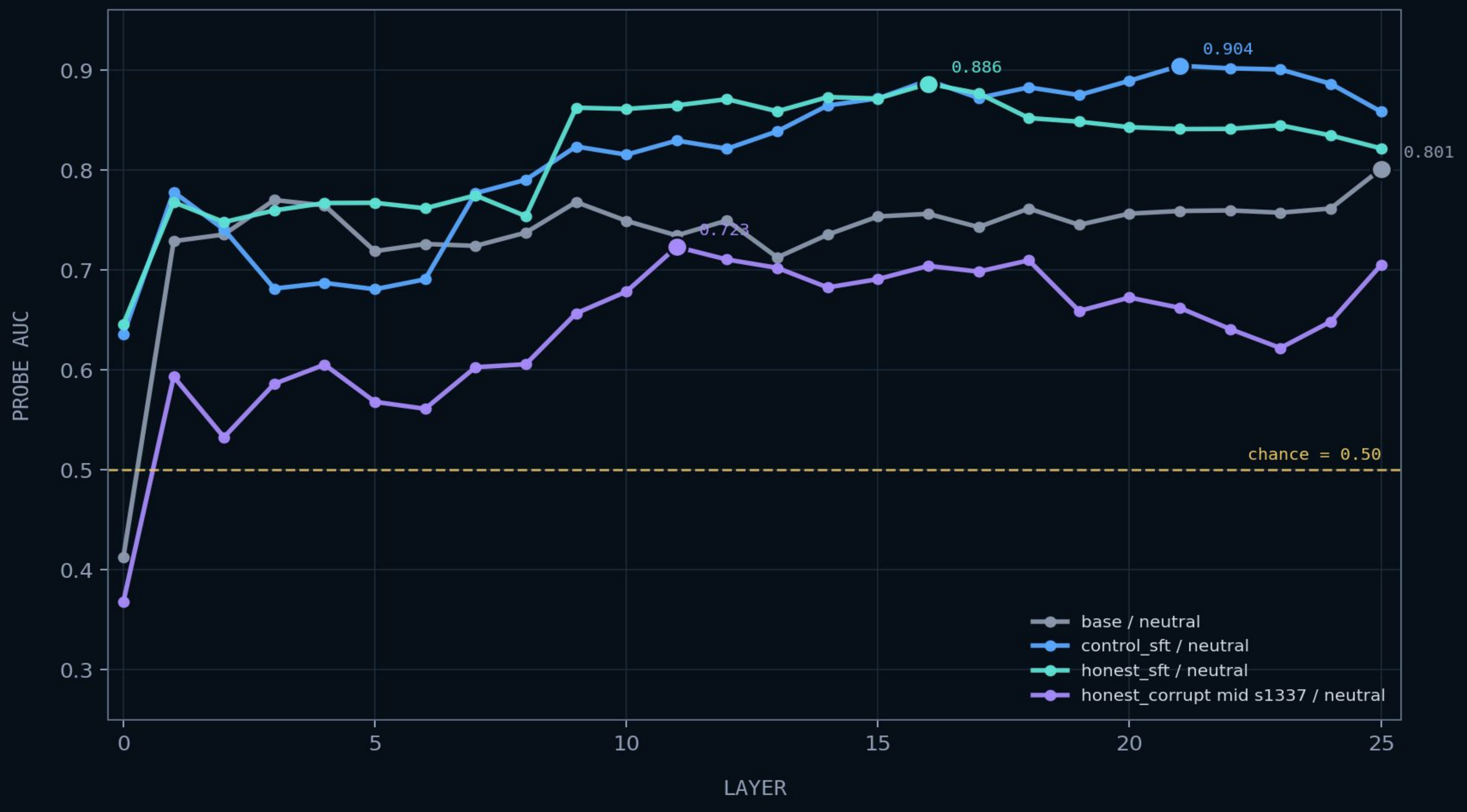
0.978
HIGH · control

1.000
HIGH · honest

Probe trainability gate (min(pos,neg) ≥ 5) flips from PASS at low/mid → FAIL at high in 14 of 18 corruption cells.

Activation probes recover signal above chance

This is the weak bar: decodable activation signal exists, but it may still reflect artifacts in the examples.



SOURCE · mid-dose probes · within_cell_kfold5 · selected cells shown

18/18

mid-dose cells were probe-trainable after reducing corruption pressure

0.61-0.90

best-layer probe separability range across the 18 cells

weak gate

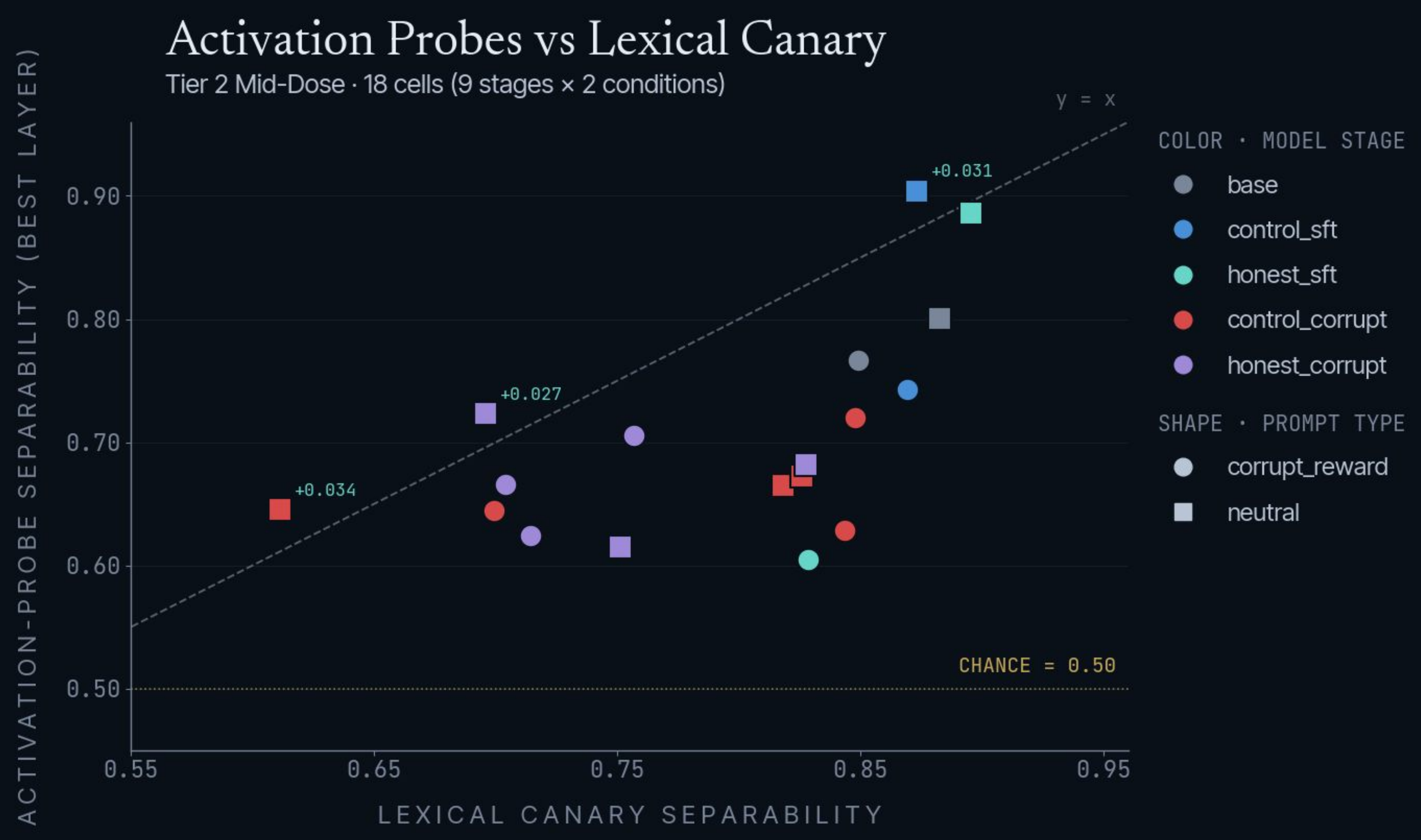
above-chance probes do not by themselves prove an internal deception feature

TAKEAWAY

Probe trainability is real enough to analyze, but not enough to claim a clean detector.

Activation probes ≤ prompt-only canary in 15 of 18 mid-dose cells

Logistic regression on residual-stream activations · 5-fold grouped CV · best layer per cell. Plot separability as max(AUC, 1–AUC); class-flip is real and we keep it visible.



SOURCE · probe_vs_lexical_summary.csv · 18 mid-dose cells

Points above the y=x line indicate cells where activation probes beat the lexical canary. 3 of 18 do (small positive deltas).

METRIC	CELLS	BEST LAYER	Δ vs CANARY
trainable	18/18	-	-
probe > prompt-only	3/18	L4/L21/L11	+.027..+.034
prompt-only ≥ probe	15/18	-	-
best positive cell	s42*	L4	+.034

Deception search:
3 cells scattered across 3 layers, no coherent depth-localization claim

TAKEAWAY

Trainable probes exist, but only 3 of 18 cells beat the prompt-only canary, and only weakly. No internal lie detector claim.

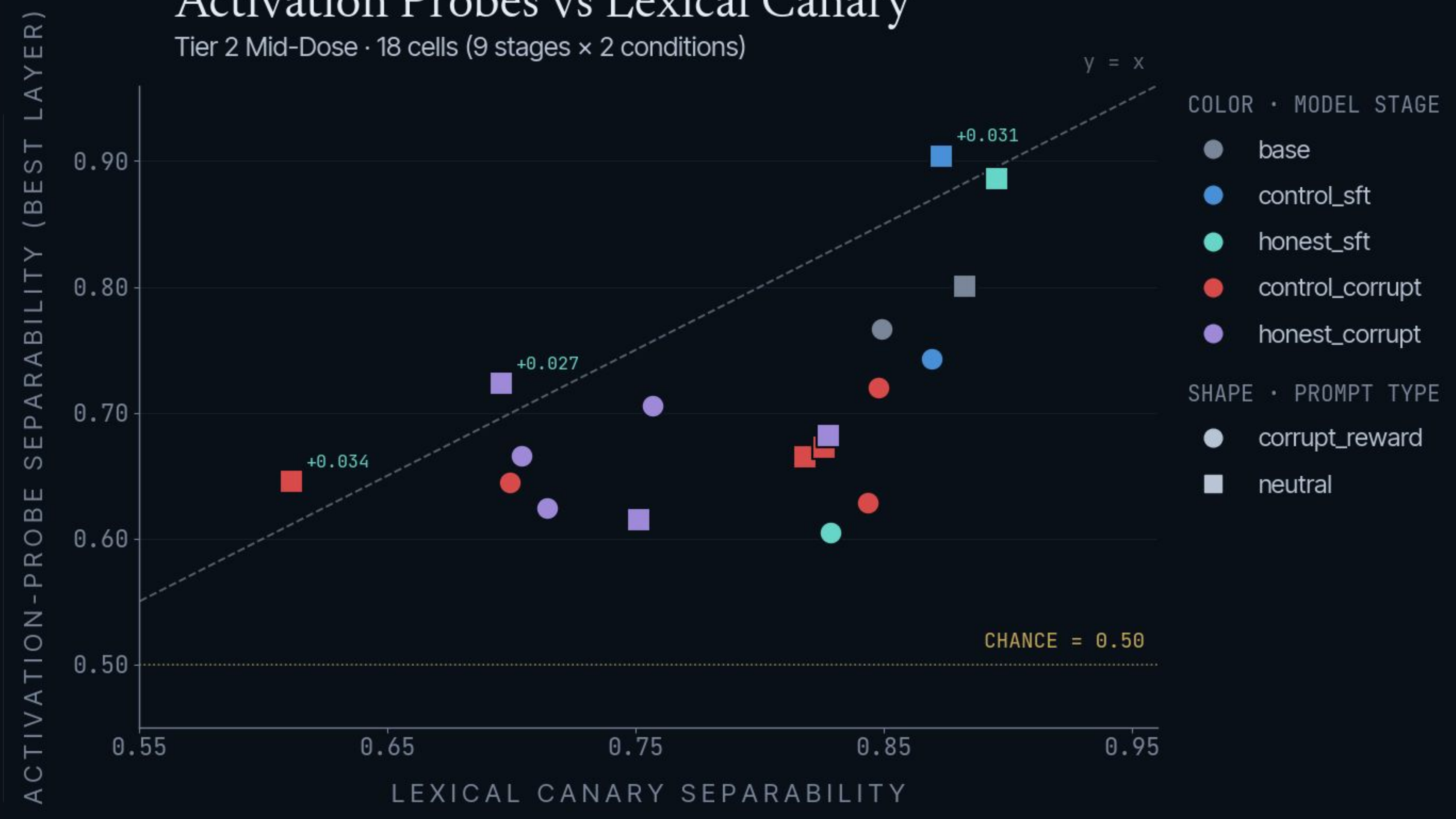
* - 3 test cells beat canary:c_corrupt s42·neutral (+.034), c_sft·neutral (+.031), h_corrupt s1337·neutral (+.027)

Prompt-only canary explains the probe verdict

Same logistic regression, different features – bag-of-words on the user-visible prompt only. Wins 15 of 18 cells. Provides the floor that activation probes must beat to claim a deception detector.

Activation Probes vs Lexical Canary

Tier 2 Mid-Dose · 18 cells (9 stages × 2 conditions)



SOURCE · probe_vs_lexical_summary.csv · 18 mid-dose cells

Points above the y=x line indicate cells where activation probes beat the lexical canary. 3 of 18 do (small positive deltas).

FINDING	BASIS	VALUE
prompt-only wins	15/18 cells	canary floor
best prompt AUC	honest_sft	0.907
backend portability	MPS vs CUDA	~0.02 drift
behavior drift	MPS vs CUDA	~0.20 drift

Methodology anchor:
Control-task probing discipline (Hewitt & Liang 2019; Belinkov, Voita, Pimentel)

TAKEAWAY

Prompt-only features are backend-stable (~0.02 drift) while behavior moves ~0.20. This is the floor activation probes must clear.

Strict gate caught a CUDA generation drift

Strict parsed-label equivalence on 360-row subset – base / control_sft / honest_sft × reward-incentive / neutral × 60 rounds. Variance-rich stages, not saturated corruption.

BACKEND	OUTCOME
MPS / fp32	PASS at batches 4 / 8 / 16 – 3.77× speedup at batch 16 over 600s serial (159s, RSS 9.37 GB). Canonical.
CUDA non-quantized	FAIL at base/shell/reward-incentive seed 3 – claimed_position serial=2 vs batched=0. Generation determinism break, not parser.
CUDA 4-bit	Not tested – gate failed at non-quantized layer first. Correct gate behavior; saved compute.
Outcome	Canonical results stay on MPS/fp32. GCP CUDA/4-bit kept as backend-variant ablation, not promotable.

3.77×
BEST SPEEDUP
MPS batch 16 vs serial

PASS
MPS BATCHES
4 / 8 / 16 all equivalent

FAIL
CUDA NON-Q
claimed_position drift

GATE
METHODOLOGY ANCHOR
pairs with prompt-only canary

Transfer smoke test, not a broad claim

One Python-intent prompt; regex-only detector; no tool execution. Diagnostic only.

0 / 9

STAGES FLAGGED · ONE-PROMPT SMOKE TEST

1 PROMPT COUNT single intent smoke test	regex DETECTOR over output text	none TOOL EXECUTION no sandbox call yet	9/9 STAGES TESTED all returned 0

TAKEAWAY

No regex-matched tamper observed on the current single prompt.

Not the same as “no transfer exists.” The transfer suite needs more prompts and actual tool execution before it can carry a generalization claim.

The contribution is a reliable measurement harness

01

Behavior induction is strong and reproducible at high dose.

Three seeds × control and honest parents × 500 deceptive rows × 3 epochs reaches ≥0.978 deception across both prompt types.

02

Mid-dose restores the label variation probes need.

Mid-dose (250 Deceptive + 250 Honest · 0.20 epoch) lifts deception ~0.10–0.13 above parents while keeping all 18 corruption probe cells trainable.

03

Probe discipline prevents an inflated detector claim.

Prompt-only canary still wins 15 of 18 cells. We report separability, not raw AUC, and document class-flips. Activation probes ≤ prompt-only except in 3 cells.

04

Backend equivalence prevents mixing non-equivalent runs.

Strict gate caught a CUDA generation-determinism break at base/reward-incentive; canonical results stay on MPS/fp32 (batch 16, 3.77× speedup).

Honest systems-and-methods result for May 12

How the deck content lands in the ACM-format final report. Framing as measurement contribution, not an internal-detector claim.

SECTION	CONTENT
Data	Prompt-clean v2 corpus · 1k SFT rows · scored transcript labels · grouped train/test splits
Methods	LoRA SFT doses (low / mid / high) · shell-game eval · linear probes · prompt-only canary · equivalence gate
System	Artifact-backed repo pipeline · sha256-anchored evidence trees · single-command node reproduction
Experiments	high-dose ceiling · mid-dose probe-testable · GCP backend variant · batch-equivalence MPS/CUDA

4	1	5	0
MAIN FIGURES	MAIN TABLE	CONTRIBUTIONS	DEFENSIVE CLAIMS
F1 pipeline · F2 dose · F3 probe-vs-prompt-only · F4 trainability	T1 per-stage summary	i–v from §1 Introduction	no internal lie detector claimed

Team RAV.ai · Questions?

Reproducible reward-gaming testbed · dose-response characterization · probe-trainability window · out-of-incentive generalization.

WHAT'S NEXT	May 12 paper: bootstrap CIs · expanded probes · held-out family
OPEN PROBLEMS	Activation steering at best probe layer (causal upgrade)
STILL PENDING	CUDA equivalence-gate debug for transfer beyond MPS
SCOPE GUARD	“Misreporting” / “reward gaming” unless an operational label justifies “deception”

FINAL THOUGHTS

Don't scale the model to hide an eval problem.

TEAM RAV.AI – QUESTIONS?

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Alexander Xin (Wei)	PROBES
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